Task 1 - Creating a list

Description

Just create a list of any size with any values.

Lets discuss what is a list. There are two points of view - conceptual and from a language perspective.

Conceptualy a list is a data structure, in which you can store some values in some order.

From a python perspective we can describe list in a different ways. List is a type like int or string. Also list is an iterable - that means we can iterate through it.

The fact that something is iterable means we can do like this:

```
for i in iterable:
```

String is iterable by the way, range() is also an iterable.

There are several ways to create list, i will not explain them here, just will leave a list of them

someList = [] - empty list

someList = list() - also empty

someList = [1, 2, 3] - list with initial values

someList = [“4”] * 10 - list containing ten “4” values

someList = [i for i in range(10)] - this one is called list comprehension, very useful thing. Try to guess what will be contained in created list.

Task 2 - Adding to a list

Description

Create an emty list and add values from 0 to 10.

Output Data Format

A list containing numbers from 0 to 10

Examples

Input: none

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

Try add a * before list variable name when printing it. This is called list unpacking

Task 3 - Squaring

Description

You are given a list and an index. Print a square of number that is stored in list by given index.

Input Data Format

A list of numbers, and an integer index value

Output Data Format

Integer number

Examples

```
Input:
2 3 1 4
3
```

```
16
```

Task 4 - Reversing

Description

You are given a list, reverse it. (you can't use list.reverse() method).

Input Data Format

A list.

Output Data Format

Reversed list.

Examples

```
Input:
1 2 3 4 5 6
```

```
6 5 4 3 2 1
```

For practice purpuses you can try out list.reverse() method. Also you can google and look for how is this method done, is it different from your solution?

Task 5 - Poisoning enemies

Description

There are 10 enemies, each one has 10 health point initially.
Enemy can become poisoned. Being poisoned means an enemy loses 1 hp every 1 second.
Given a time in seconds and some additional commands calculate enemies hp values by the end of time.

Input Data Format

Integer S - time in seconds
Every second you receive a command - Integer n - number of enemy.
If this enemy is not poisoned it becomes poisoned.
If this enemy is poisoned it becomes normal.
If command −1 is given - nothing happens.

Output Data Format

A list of enemies health points.

Examples

```
Input:
5
1
2
2
-1
4
```

```
10 5 9 10 9 10 10 10 10 10
```

Task 6 - Sorting

Description

You are given a list of numbers in arbitrary order. Sort list so it contains values in descending order. (again, you can’t use list.sort() method).

There a lot of sorting algorithms - choose one that you will find easy and interesting enough. My suggestion for first choice is a bubble sort or insertion sort.

Input Data Format

A list of numbers.

Output Data Format

A list of numbers.

Examples

```
Input:
2 3 1 4 8 2
```

```
8 4 3 2 2 1
```

Task 7 - Higher dimension

Description

Did you know you can store a list inside a list? Well - you can, now try it out and print a result.

Two dimensional list is also called a table or matrix. Create a table of 4 by 3 size (4 - number of rows, 3 - number of columns) and fill it with zeros.

Now iterate through it and change every value to a sum of its indices.

Output Data Format

A table.

Examples

Input: none

```
0 1 2
1 2 3
2 3 4
3 4 5
```

Notice that you should print it out not like [[1, 2, 3], [2, 3, 4]] this.

Task 8 - Matrix addition

Description

Now you have to implement a matrix addition logic. Having two matrixes of same sizes you can perform an addition operation by calculating a sum between values of same positions.

Matrixes are big, it is not very handy to type them in terminal for testing your program - so lets read them from file.

Having a file named input.txt you can open it like this:

```
with open(“input.txt”, “r”, encoding=“utf-8”) as file:
  fileContent = file.read()
```

There are other file methods to read from file and write to it, you can look them up in python documentation.

Input Data Format

Two numbers - sizes of matixes. Two matrixes of give sizes.
Inside a file input.txt

Output Data Format

A matrix inside a file output.txt

Examples

```
Input input.txt:
2 2

1 1
1 1

4 4
5 5
```

```
5 5
6 6
```

Try to make matrix1 and matrix2 stay unchanged, explore how to copy a normal list and nested list. Try different ways and print all matrixes to see the results.